Holding time and exits

"Spread captured" is only real if you can get out at fair value, and you can't: the exit is a second trade with its own cost. Holding longer buys a cheaper exit and costs risk that grows with the square root of time, so the question is which of the two wins on your coins.

All numbers below come from the synthetic simulation in `mm_economics_vs_frequency.ipynb` (80% vol, base-tier fees, 750 ± 100 ms to Hyperliquid). Read the shapes, not the levels.

Spread captured assumes a free exit

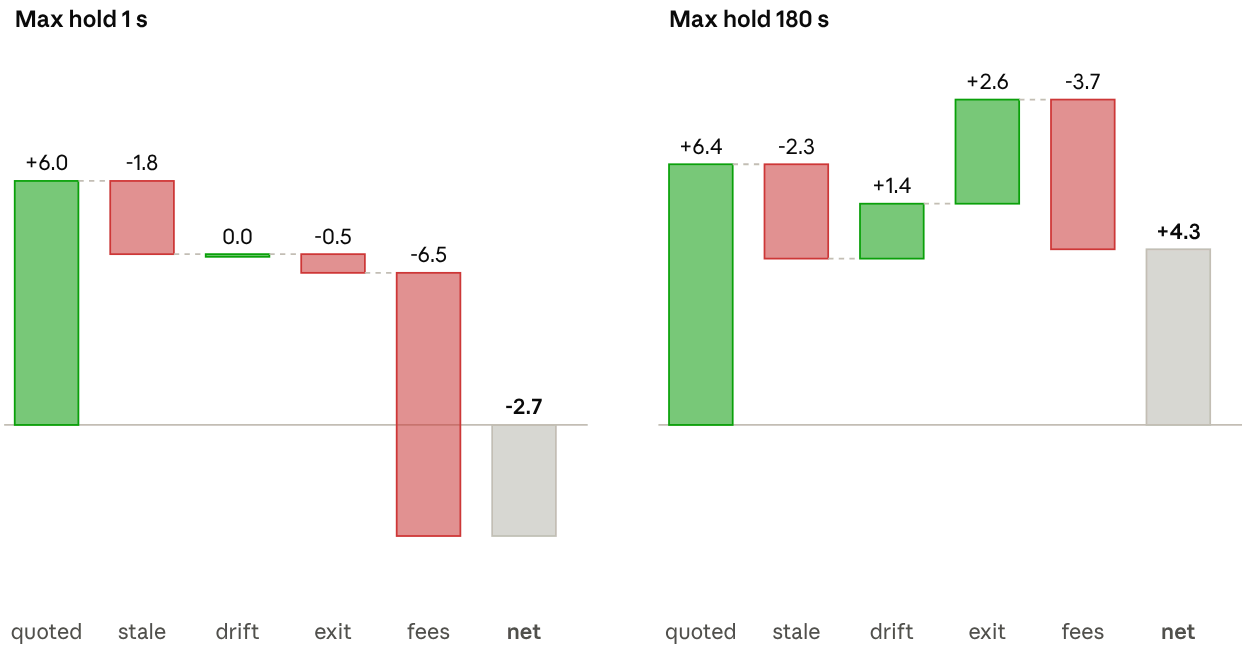
The five-term breakdown measures spread against fair value at the moment of the fill. The exit is a second trade, so the honest per-lot identity adds it explicitly. Here d = +1 for a long lot, F is the Binance mid, and F^q is the fair you used when you sent the quote:

```
\text{P\&L}_{\text{lot}} = \underbrace{d(F^{q} -
p_{\text{in}})}_{\text{quoted edge}} + \underbrace{d(F_{\text{fill}} -
F^{q})}_{\text{staleness}} + \underbrace{d(F_{\text{exit}} -
F_{\text{fill}})}_{\text{holding drift}} + \underbrace{d(p_{\text{out}} -
F_{\text{exit}})}_{\text{exit vs fair}} - \text{fees}
```

The terms telescope, so they always add up to the realised P&L.

Holding up to 180 s turns a losing lot into a winning one, mostly through a cheaper exit

bp per lot. Green adds to P&L, red takes away, grey is the net. Same quotes and fees; only the maximum holding time changes.



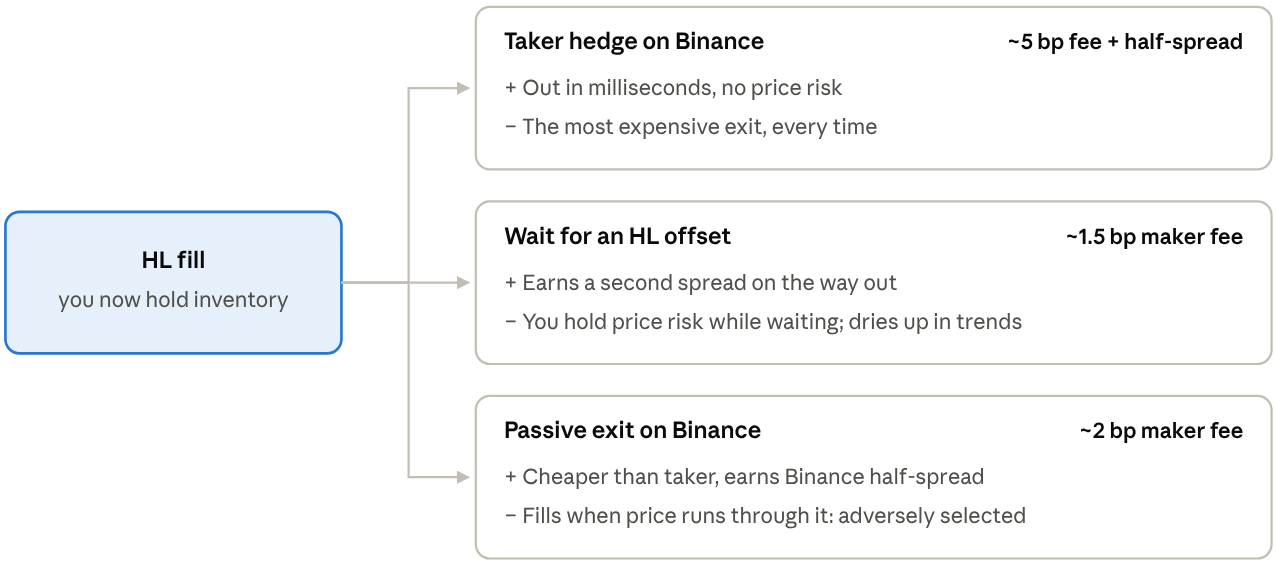
mm_economics_vs_frequency.ipynb · Monte Carlo, 3 seeds × 2 h each, 6 bp half-spread

Quoted edge and staleness barely move between the two panels. The difference is almost all in the last two terms: at 180 s about 80% of lots close against an offsetting Hyperliquid fill, which earns a spread back and avoids the 5 bp Binance taker fee.

Three ways out

Every exit trades certainty against cost. The cheap ones make you wait, and waiting is when the market decides whether you were right.

Three ways to get out of a Hyperliquid fill



fees at base tier, from the main tab

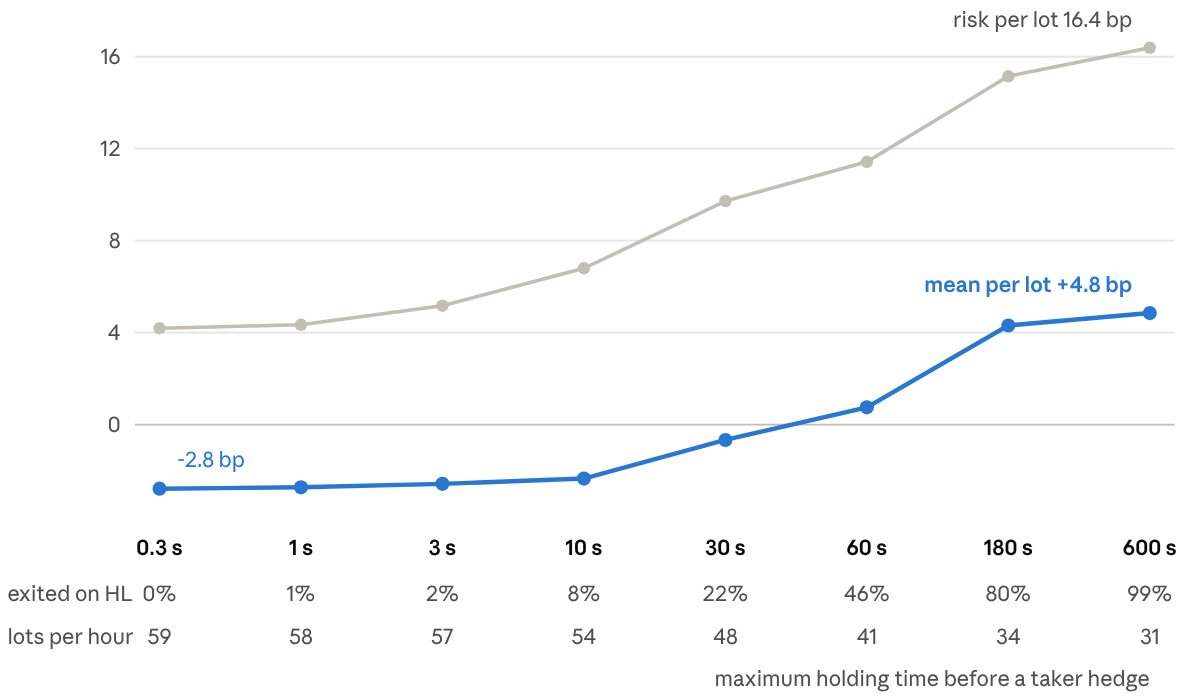
The passive exits are the subtle trap. A resting exit order fills when price moves through it, which is exactly when you'd rather it hadn't. Getting out "at fair" passively is adversely selected just like getting in.

Holding longer: a cheaper exit, bought with risk

Three things move together as you hold longer, and only one of them helps.

Edge per lot levels off near +5 bp while risk per lot keeps climbing

bp per lot against maximum holding time. Risk is the standard deviation of one lot's P&L.



	0.3 s	1 s	3 s	10 s	30 s	60 s	180 s	600 s
exited on HL	0%	1%	2%	8%	22%	46%	80%	99%
lots per hour	59	58	57	54	48	41	34	31

maximum holding time before a taker hedge

mm_economics_vs_frequency.ipynb · holding sweep, 3 seeds × 2 h each

- **The helpful one: the exit gets cheaper.** The share of lots closed by an offsetting Hyperliquid fill rises from almost none to nearly all, and each of those saves the taker fee and earns a spread.
- **Risk grows like the square root of time.** At 80% vol a lot's price risk is about 1.4 bp at 1 s, 11 bp at 60 s and 25 bp at 5 minutes, while the edge it buys stays in single digits.
- **Throughput falls.** With an inventory cap, open lots ≈ lot rate × holding time, so longer holds mean fewer lots per hour: 59 down to 31 here.

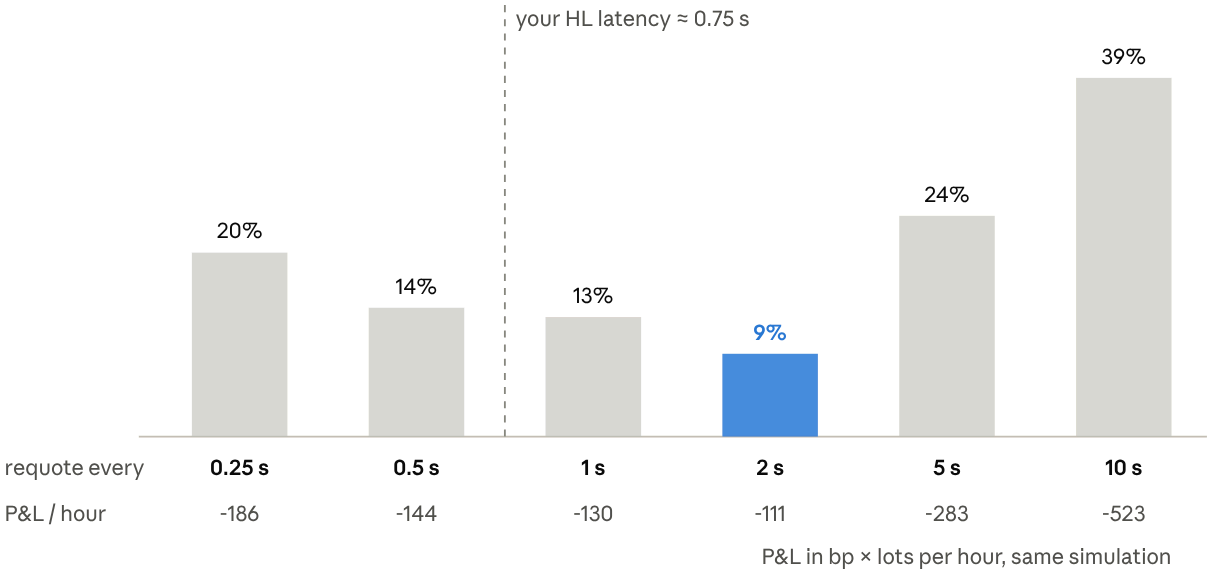
Past a few minutes you are no longer market making. You are holding a directional position, and the drift term, not the spread, decides your P&L.

Requoting faster than your latency doesn't help

The other frequency dial is how often you re-centre your quotes, and it is not monotone.

Requoting about every 2 s gets picked off least; faster or slower both cost you

Share of Hyperliquid fills that were informed, at a 6 bp half-spread, against how often you re-centre your quotes.



P&L in bp × lots per hour, same simulation

mm_economics_vs_frequency.ipynb · requote × half-spread grid, 6 bp row, 2 seeds × 1 h each

- **Too slow (5–10 s):** quotes sit at prices Binance has already left, and arbs take them.
- **Too fast (sub-second):** every update still lands about 750 ms late, so you keep re-posting stale quotes into a market that is still moving. In a run, that puts a fresh target in front of the arb several times.
- Width matters more than either: in the full grid in the notebook, nothing below about an 8 bp half-spread is profitable at these parameters, whatever the requote interval.

What this means, and what to measure

The Kearns link. Holding time is where Kearns' $(K - z^2)/2$ bites. The longer you hold, the more z^2 (net move against your inventory) you are exposed to, and only mean reversion keeps it bounded. If you hedge on Binance you carry the basis, which does mean-revert; if you wait for an HL offset you carry the price, which at seconds-to-minutes is close to a random walk.

Read the holding result with suspicion. It is the most flattering number in the notebook, for two reasons:

- The simulated uninformed flow arrives symmetrically whatever the price is doing, so an offsetting fill always turns up. Real flow trends, and the side you need dries up exactly when you are holding the wrong way.

- The +1.4 bp holding drift at 180 s should be zero on a random walk; it is simulation noise and inventory skew, not edge. A 15 bp risk per lot needs many more lots than 3 seeds × 2 hours to pin down a few bp of mean.

What to measure from your data, in order:

- [ ] Offset rate conditional on the recent Binance move: how often an opposite Hyperliquid fill arrives within 10, 60 and 180 s after a fill, split by whether Binance moved with you or against you.
- [ ] The markout curve of real Hyperliquid fills against the Binance mid, 0–180 s. Where it sits at each holding time is your realised spread before fees.
- [ ] Basis half-life (HL mid minus Binance mid). If it is much shorter than your holding time, the hedge-and-carry-basis route is the one Kearns favours.
- [ ] Your firm's real fee tier: every one of these curves moves with the exit fee.